

Experiment No-4

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Aim of the practical:

To understand and implement different types of SQL Joins including INNER JOIN, LEFT JOIN, RIGHT JOIN, and FULL JOIN, CROSS JOIN to combine data from multiple tables in a relational database.

Objective of the practical:

To gain the knowledge and practical skills required to use various SQL JOIN operations (INNER, LEFT, RIGHT, FULL, CROSS) in order to retrieve and combine related data from multiple tables in a relational

Steps to Implement SQL Joins:

Step1:

Create Database and Two Tables and Insert Data.

Step2:

Write the query of INNER JOIN.

Step3:

Write the query of LEFT JOIN.

Step4:

Write the query of RIGHT JOIN.

Step5:

Write the query of FULL JOIN.

Step6:

Write the query of CROSS JOIN.

Solution:

1. Create Database and Two Tables and Insert Data

```

1 • create database BA;
2 • use BA;
3 • create table Customers(
4     customerID int primary key,
5     CustomerName varchar (50),
6     Country varchar (50)
7 );
8
9 • insert into Customers(customerID, customerName, country) values
10 ('1', 'Harsh', 'India'),
11 ('2', 'Kanu', 'Canada'),
12 ('3', 'Pooja', 'UK'),
13 ('4', 'Rahul', 'Norway');
14
15 • create table Orders(
16     OrderID int,
17     CustomerID int ,
18     OrderDate date );
19
20 • insert into Orders(orderID, customerID, OrderDate)values
21 ('101', '1', '2023-06-15'),
22 ('102', '2', '2023-09-25'),
23 ('103', '3', '2024-05-19'),
24 ('104', '4', '2020-07-25');

```

2. INNER JOIN

Returns records that have matching values in both tables.

```

26
27 • SELECT * FROM customers
28     INNER JOIN orders
29     ON customers.customerID = orders.customerID;
30

```

	customerID	CustomerName	Country	OrderID	CustomerID	OrderDate
▶	1	Harsh	India	101	1	2023-06-15
	2	Kanu	Canada	102	2	2023-09-25
	3	Pooja	UK	103	3	2024-05-19
	4	Rahul	Norway	104	4	2020-07-25

3. LEFT JOIN (or LEFT OUTER JOIN)

Return all records from the left table, and the matched records from the right table.

```
select Customers.customerID, Customers.country, Orders.OrderID, Orders.OrderDate
from Customers
left join Orders
on Customers.customerID = Orders.orderid
```

	customerID	CustomerName	Country	OrderID	CustomerID	OrderDate
▶	1	Harsh	India	151	1	2023-06-15
	2	Kanu	Canada	152	2	2023-09-25
	2	Kanu	Canada	153	2	2024-05-19
	1	Harsh	India	154	1	2024-01-20
	3	Pooja	UK	155	3	2025-07-19

4. RIGHT JOIN (or RIGHT OUTER JOIN)

Returns all records from the right table, and the matched records from the left table.

```
39 select Customers.customerID, Customers.country, Orders.OrderID, Orders.OrderDate
40 from Customers
41 right join Orders
42 on Customers.customerID = Orders.orderid;
```

	customerID	country	OrderID	OrderDate
▶	1	India	NULL	NULL
	2	Canada	NULL	NULL
	3	UK	NULL	NULL
	4	Norway	NULL	NULL
	NULL	NULL	101	2023-06-15
	NULL	NULL	102	2023-09-25
	NULL	NULL	103	2024-05-19
	NULL	NULL	104	2020-07-25

5. FULL JOIN (or FULL OUTER JOIN)

Returns all records when there is a match in either left or right table

```
32 • select Customers.customerID, Customers.country, Orders.OrderID, Orders.OrderDate
33 from Customers
34 left join Orders
35 on Customers.customerID = Orders.orderid
36
37 union
38
39 select Customers.customerID, Customers.country, Orders.OrderID, Orders.OrderDate
40 from Customers
41 right join Orders
42 on Customers.customerID = Orders.orderid;
```

	customerID	country	OrderID	OrderDate
▶	1	India	NULL	NULL
	2	Canada	NULL	NULL
	3	UK	NULL	NULL
	4	Norway	NULL	NULL
	NULL	NULL	101	2023-06-15
	NULL	NULL	102	2023-09-25
	NULL	NULL	103	2024-05-19
	NULL	NULL	104	2020-07-25

6. CROSS JOIN

Returns the Cartesian product of the two tables (all possible combinations).

```

32 • select Customers.customerID, Customers.country, Orders.OrderID, Orders.OrderDate
33 from Customers
34 left join Orders
35 on Customers.customerID = Orders.orderid
36

```

	customerID	country	orderID	orderDate
▶	4	Norway	101	2023-06-15
	3	UK	101	2023-06-15
	2	Canada	101	2023-06-15
	1	India	101	2023-06-15
	4	Norway	102	2023-09-25
	3	UK	102	2023-09-25
	2	Canada	102	2023-09-25
	1	India	102	2023-09-25
	4	Norway	103	2024-05-19
	3	UK	103	2024-05-19
	2	Canada	103	2024-05-19
	1	India	103	2024-05-19
	4	Norway	104	2020-07-25
	3	UK	104	2020-07-25
	2	Canada	104	2020-07-25
	1	India	104	2020-07-25

Learning Outcomes:

1. Understood the purpose of SQL JOINS and when they are needed in relational databases.
2. Understood difference between INNER JOIN, LEFT JOIN, RIGHT JOIN, and FULL OUTER JOIN and describe their behavior.
3. Understood SQL queries using different JOIN types to combine data from multiple tables.